

UNIT 15

CURRENT ELECTRICITY



ELECTRIC CURRENT

Electric current is the flow of electric charge through a conductor, such as a wire, due to the movement of electrons or other charge carriers.

CONVENTIONAL CURRENT

Conventional current is the hypothetical direction of positive charge flow, opposite to the actual flow of electrons. It is used for ease of understanding and calculation in circuit analysis.

$$I = \frac{q}{t}$$

TYPES OF CURRENT

There are two types of currents,

Direct Current (DC): Direct current is an electric current that flows in one direction only, maintaining a constant polarity over time, commonly produced by batteries or DC power sources.

Alternating Current (AC): Alternating current is an electric current that reverses direction periodically, oscillating between positive and negative values, typically used in household electricity supply and generated by alternating voltage sources.

POTENTIAL DIFFERENCE

Potential difference, also known as voltage, is the electric potential energy per unit charge between two points in an electric circuit. It drives the flow of electric current.

FORMULA

The potential difference (V) between two points is calculated using the formula:

$$V = \frac{W}{Q}$$

Where:

- (V) = Potential difference (measured in volts, V)
- (W) = Work done in moving charge between the points (measured in joules, J)
- (Q) = Charge moved between the points (measured in coulombs, C)

S.I Unit:

The SI unit of potential difference is the Volt (V).

ELECTROMOTIVE FORCE (EMF)

Electromotive force is the energy per unit charge supplied by a source, such as a battery or generator, to drive electric current in a circuit. It's not a force in the traditional sense but rather a potential difference.

Formula: The electromotive force (E) is related to the potential difference (V) across the terminals of a source and the internal resistance (r) of the source:

$$E = \frac{\text{Energy Supplies (W)}}{\text{Unit Charge (q)}}$$

Where:

- (E) = Electromotive force (Volts, V)
- (V) = Potential difference (Volts, V)
- (I) = Current flowing through the circuit (Amperes, A)
- (r) = Internal resistance of the source (Ohms, Ω)

S.I Unit: SI unit of EMF is Volt (V).

OHM'S LAW

Ohm's Law states that the current flowing through a conductor is directly proportional to the voltage across it, at a constant temperature.

FORMULA

Ohm's Law is mathematically represented as:

$$I = \frac{V}{R}$$

Where:

- (I) = Current flowing through the conductor (Amperes, A)
- (V) = Voltage across the conductor (Volts, V)
- (R) = Resistance of the conductor (Ohms, Ω)

LIMITATIONS OF OHM'S LAW

1. Temperature Dependency: Ohm's Law assumes constant temperature, while resistance can change with temperature.

2. Non-Ohmic Materials: It's not applicable to materials that don't follow a linear V-I relationship.

3. Time Dependency: It doesn't consider time-varying circuits or transient effects.

4. Low Voltage: At very low voltages, contact resistance and nonlinearities can affect accuracy.

5. Complex Circuits: Not suitable for complex circuits with nonlinear elements.

6. Frequency Effects: Not valid for high-frequency AC circuits due to capacitance and inductance effects.

7. Nonlinear Elements: Doesn't account for components like diodes or transistors.

8. Nonuniform Materials: Not applicable to nonuniform materials where resistance varies along the length.

RESISTANCE

Resistance is the opposition offered by a material to the flow of electric current. It is caused by collisions between electrons and atoms in the material, converting electrical energy into heat.

$$R = \frac{V}{I}$$

FACTORS AFFECTING RESISTANCE

1. Length: Longer conductors have higher resistance.

2. Cross-Sectional Area: Thinner conductors offer more resistance.

3. Temperature: Higher temperatures increase resistance in most materials.

4. Material: Different materials have varying resistivity values.

USES OF RESISTANCE

1. Heating Elements: Resistance is used to generate heat in devices like toasters, heaters, and incandescent light bulbs.

2. Voltage Division: Resistors are used to divide voltage in circuits, enabling specific voltage levels.

3. Current Limitation: Resistors control current flow and prevent damage in electronic circuits.

4. Signal Conditioning: They're used in electronics to modify signal amplitudes and shapes.

5. Circuit Protection: Resistors in fuses and surge protectors prevent excessive current flow.

6. Sensors: Certain resistive materials change resistance with temperature, light, or pressure, making them useful in sensors.

SERIES AND PARALLEL COMBINATIONS OF RESISTORS

The method of connect the electric components is called circuit. There are two types of circuits,

- 1) Series Combinations Circuit
- 2) Parallel Combinations Circuit

SERIES COMBINATION OF RESISTORS

Ohm's Law in a series combination of resistors, resistors are connected end to end, so the same current flows through each resistor.

FORMULA DERIVATION

Let suppose three resistors R_1 , R_2 , and R_3 are connected in series. When this combination is connected to a battery of V volts, it draws current I from the battery. R_1 is a single resistor. This resistor is such that when it is connected to the same battery of V volts, it also draws current from the battery. By applying Ohm's law to each resistor, we have:

$$\cdot V_1 = IR_1$$

$$\cdot V_2 = IR_2$$

$$\cdot V_3 = IR_3$$

$$\cdot V = IR$$

Using them in equation we get:

$$\cdot IR = IR_1 + IR_2 + IR_3$$

$$\cdot IR = I(R_1 + R_2 + R_3)$$

$$\cdot R = R_1 + R_2 + R_3$$

ADVANTAGES

1. Easy to Calculate: Total resistance is the sum of individual resistances.

2. Voltage Division: Voltage is divided proportionally among resistors, useful for circuits requiring specific voltage drops.

DISADVANTAGES

1. High Total Resistance: Total resistance increases with each resistor, reducing overall current flow.

2. Equal Current: The same current flows through each resistor, limiting flexibility in current distribution.

Series combinations are useful when specific voltage drops are required or when using multiple resistors to achieve a desired total resistance. However, the increase in total resistance can impact overall circuit performance.

PARALLEL COMBINATION OF RESISTORS

In a parallel combination of resistors, the resistors are connected across the same two points, allowing the same voltage to be applied across each resistor.

FORMULA

Let suppose three resistors R_1 , R_2 , and R_3 are connected in parallel. When this combination is connected to a battery of V volts, it draws current I from the battery. R_e is a single resistor. This resistor is such that when it is connected to the same battery of V volts, it also

draws current I from the battery. This resistor is therefore called the equivalent resistor and its resistance is called equivalent resistance.

- $I = V / R$

where V is the voltage across the circuit and R is the equivalent resistance.

Applying Ohm's law to each resistor, we have:

- $V_1 = IR_1$

- $V_2 = IR_2$

- $V_3 = IR_3$

where V_1 , V_2 , and V_3 are the voltages across resistors R_1 , R_2 , and R_3 , respectively.

Summing these equations, we get:

- $V_1 + V_2 + V_3 = I(R_1 + R_2 + R_3)$

or

- $V = IR_1 + IR_2 + IR_3$

Dividing both sides by V , we get:

- $I = 1/R_1 + 1/R_2 + 1/R_3$

or

- $R = 1/R_e = (1/R_1 + 1/R_2 + 1/R_3)$

This shows that the equivalent resistance of a resistor is equal to the sum of the reciprocals of individual resistances.

ADVANTAGES

1. Low Total Resistance: Total resistance decreases with each resistor, enhancing overall current flow.

2. Voltage across Each Resistor: All resistors share the same voltage, allowing independent current paths.

DISADVANTAGES

1. Complex Calculations: Calculating total resistance involves reciprocal terms.

2. Voltage Division: Voltage is not equally divided among resistors, which can lead to unequal power dissipation.

Parallel combinations are useful when maintaining voltage consistency is important or when multiple resistors need to achieve a lower total resistance. However, complex calculations and potential variations in voltage distribution should be considered.

ELECTRIC POWER AND JOULE'S LAW

ELECTRIC POWER

Electric Power is the rate at which electrical energy is transferred by an electric circuit. It is measured in watts (W), which is equal to joules per second (J/s).

POWER DISSIPATION

Power Dissipation is the rate at which electrical energy is converted into heat by an electrical device. It is also measured in watts (W).

$$P = VI$$

Where:

P is the electric power in watts (W)

V is the voltage in volts (V)

I is the current in amperes (A)

ENERGY IN A RESISTOR

Energy in a resistor is the electrical energy that is converted into heat by the resistor. The electrical energy that is converted into heat by a resistor is equal to the product of the current that flows through the resistor, the voltage drop across the resistor, and the time for which the current flows.

FORMULA

The formula for the energy in a resistor is:

$$E = I^2 R t$$

Where:

E is the energy in joules (J)

I is the current in amperes (A)

R is the resistance in ohms (Ω)

t is the time in seconds (s)

JOULE'S LAW

Joule's Law states that the amount of heat generated in a conductor is proportional to the square of the current flowing through the conductor, the resistance of the conductor, and the time for which the current flows.

FORMULA

The formula for Joule's Law is:

$$H = I^2 R t$$

Where:

H is the heat energy in joules (J)

I is the current in amperes (A)

R is the resistance in ohms (Ω)

t is the time in seconds (s)

USES OF CIRCUIT COMPONENT

Diodes: Diodes are used to allow current to flow in one direction only. They can be used to create rectifiers, clippers, and switches.

Resistors: Resistors are used to control the flow of current in a circuit. They can be used to create voltage dividers, current limiters, and filters.

Capacitors: Capacitors are used to store electrical energy. They can be used to smooth out current fluctuations, create timing circuits, and filter noise.

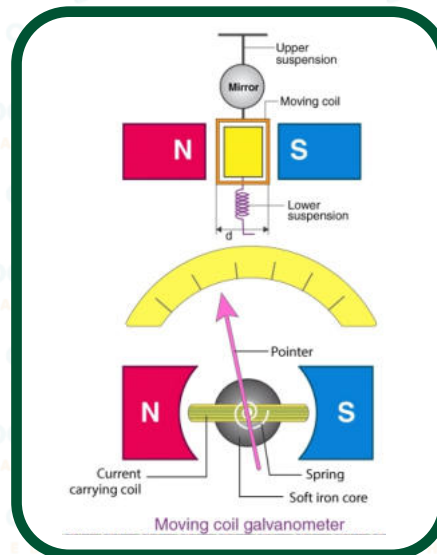
Inductors: Inductors are used to store magnetic energy. They can be used to create filters, oscillators, and transformers.

Transistors: Transistors are used to amplify or switch electronic signals. They are the basic building blocks of most modern electronic devices.

Integrated circuits: Integrated circuits (ICs) are complex circuits that are built on a single semiconductor chip. They can contain millions of transistors and other components. ICs are used in a wide variety of electronic devices, including computers, smartphones, and TVs.

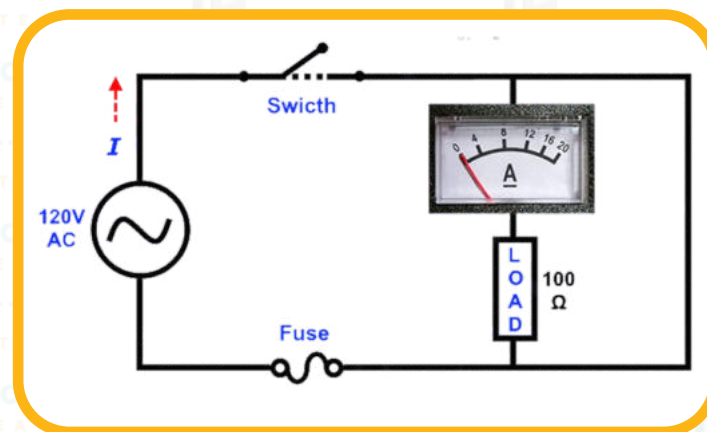
MOVING COIL GALVANOMETER

A moving coil galvanometer is a type of electrical instrument that measures electric current. It consists of a coil of wire suspended in a magnetic field. When a current flows through the coil, it generates a force that deflects the coil. The amount of deflection is proportional to the current flowing through the coil.



AMMETER

An ammeter is a type of electrical instrument that measures electric current. It is connected in series with the circuit that is being measured. The ammeter has a very low resistance, so it does not significantly affect the current in the circuit.



VOLTMETER

A voltmeter is a type of electrical instrument that measures electric potential difference. It is connected in parallel with the circuit that is being measured. The voltmeter has a very high resistance, so it does not significantly draw current from the circuit.

Here is a table that summarizes the key differences between moving coil galvanometer, ammeter, and voltmeter:



FEATURE	MOVING COIL GALVANOMETER	AMMETER	VOLTMETER
Type of instrument	Electrical	Electrical	Electrical
Measures	Electric current	Electric current	Electric potential difference
Connection to circuit	Series	Series	Parallel
Resistance	Low	Very low	Very high
Effect on circuit	Affects current	Does not affect current	Does not affect current

HAZARD OF ELECTRICITY

Electricity is a powerful force that can be dangerous if not handled properly. The hazards of electricity include electric shock, fire, and explosion.

DAMAGED INSULATION

Electrical insulation is the material that surrounds electrical wires and cables. It protects the wires from coming into contact with each other and from coming into contact with people or other objects. Damaged insulation can expose the wires and make them susceptible to electric shock.

OVERHEAD CABLES

Overhead cables are electrical cables that are suspended above the ground. They are often used to transmit electricity over long distances. Overhead cables can be a hazard if they come into contact with trees or other objects, or if they are damaged by weather or other events.

DAMP CONDITIONS

Damp conditions can increase the risk of electric shock. This is because water is a conductor of electricity. If electrical wires or cables come into contact with water, they can create a shock hazard.

SAFETY MEASURES IN HOUSEHOLD ELECTRICITY

There are a number of safety measures that can be taken to reduce the risk of electrical accidents in the home. These include:

- Using only properly insulated electrical wires and cables.
- Installing circuit breakers and fuses in the home's electrical panel.
- Keeping electrical appliances and cords away from water.
- Not overloading electrical outlets.
- Hiring a qualified electrician to make any repairs to the home's electrical system.

FUSES AND BREAKERS

Fuses and breakers are safety devices that are used to protect electrical circuits from overloads and short circuits. Fuses are designed to melt and break the circuit if the current exceeds a certain level. Breakers are designed to trip and open the circuit if the current exceeds a certain level.

The Circuit Breaker: A circuit breaker is an electrical safety device that is used to protect electrical circuits from overloads and short circuits. When a circuit breaker trips, it opens the circuit and stops the flow of electricity. Circuit breakers are typically installed in the home's electrical panel.

The Ground Wire: A ground wire is a safety wire that is connected to the earth. It is used to protect people from electric shock in the event of a fault in the electrical system. The ground wire provides a path for the current to flow to the earth, rather than through a person. Ground wires are typically installed in the home's electrical system.

EFFECT OF ELECTRIC SHOCK ON HUMAN BODY

Cardiac Arrest: Electric shock can cause the heart to stop beating. This is called cardiac arrest. Cardiac arrest is a life-threatening condition that requires immediate medical attention.

Respiratory Arrest: Electric shock can also cause respiratory arrest. Respiratory arrest is when a person stops breathing. Respiratory arrest is also a life-threatening condition that requires immediate medical attention.

Burns: Electric shock can cause burns on the skin. The severity of the burns will depend on the amount of current that flows through the body and the length of time that the current flows.

Muscle Contractions: Electric shock can cause muscle contractions. These contractions can be so strong that they can throw a person off their feet or cause them to lose control of their body.

Neurological Damage: Electric shock can cause neurological damage. This damage can range from temporary confusion to permanent disability.